`bispectrum`: Selective G-Bispectra Made Practical

Anonymous Author(s)
Affiliation
Address
email

Abstract

We present `bispectrum`, an open-source PyTorch library that computes selective G-bispectral invariants for seven group/domain pairs as drop-in differentiable `nn.Modules`. The selective bispectrum reduces invariant cost from $O(|G|^2)$ to $O(|G|)$ while remaining complete, i.e. it separates orbits for generic signals. Building the library required two theoretical contributions: (i) a selective $\mathrm{SO}(3)$ bispectrum on S^2 that achieves $O(L^2)$ coefficients—matching the information-theoretic lower bound—instead of $O(L^3)$ for the full bispectrum, and (ii) resolving a previously undocumented inversion failure for the octahedral group via bootstrap initialization followed by Levenberg–Marquardt refinement. We evaluate the library on three benchmarks—C_8 on PCam, octahedral on OrganMNIST3D, and $\mathrm{SO}(3)$ on Spherical MNIST—demonstrating numerically exact rotation invariance ($\sigma_{\mathrm{rot}} < 10^{-3}$) and the data-efficiency advantage of complete bispectral invariants over incomplete alternatives such as power spectrum and norm pooling. The selective forward pass runs in sub-millisecond time on GPU for finite groups, adding negligible overhead to standard training loops.

REVIEW: #1 — RESOLVED: The $\ell = 2$ gap is closed by the joint seed lemma (Lemma 18). Theorem 1 now states completeness without caveats.

1 Introduction

Equivariant neural networks [Cohen and Welling, 2016, Weiler and Cesa, 2019] enforce symmetry by construction, but they still require an invariant representation at the point where a prediction must no longer depend on orientation. In practice, invariant pooling remains a weak point: standard choices such as norm pooling, gate-based pooling, or max/sum aggregation are usable [Weiler and Cesa, 2019], but they do not preserve all equivariant information. The bispectrum offers a mathematically principled alternative, since for many groups it yields exact invariants with reconstruction guarantees [Kakarala, 1992].

The obstacle is practical rather than conceptual. Existing bispectral constructions are scattered across theory papers, domain-specific derivations, and small implementations, with no general-purpose PyTorch library covering the finite-group and compact-group cases relevant to modern geometric deep learning. Even for groups where selectivity reduces the coefficient count from $O(|G|^2)$ to $O(|G|)$ [Mataigne et al., 2024], a user still needs a numerically stable implementation, a consistent API, inversion routines, tests, and benchmarked runtime behavior.

This paper presents `bispectrum`, a PyTorch [Paszke et al., 2019] library for computing bispectral invariants and, where available, their inverses. The library supports cyclic, dihedral, torus, disk, sphere, and octahedral settings as differentiable `nn.Modules` intended to plug into standard ML workflows. Beyond the software contribution, the paper makes two technical contributions that arose during implementation: a selective $\mathrm{SO}(3)$ bispectrum on S^2 with $O(L^2)$ coefficients (Section 3.4),

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Table 1: Invariant nonlinearities for equivariant CNNs. The selective bispectrum is the only method that is simultaneously exact, complete, and linear-cost for finite groups.

Nonlinearity	Exact?	Complete?	Coefficients	Key limitation
Norm	Yes	No	$O(\|G\|)$	Discards phase
Gate	Yes	No	$O(2\|G\|)$	Scalar bottleneck
Fourier pointwise	No	Yes	$O(\|G\| \log \|G\|)$	Aliasing
Full bispectrum	Yes	Yes	$O(\|G\|^2)$	Quadratic cost
Sel. bispectrum	**Yes**	**Yes**	$\mathbf{O(\|G\|)}$	**(this work)**

and a resolution of the bootstrap inversion failure [Mataigne et al., 2024] for the octahedral group (Section 3.5).

Contributions.

1. **Software contribution.** A PyTorch library implementing bispectral invariants for seven group/domain pairs with a consistent `nn.Module`-style interface, GPU-optimised forward passes (<1 ms for finite groups), precomputed Clebsch–Gordan buffers, and parallel construction routines (Section 3.6).
2. **Selective $\mathrm{SO}(3)$ bispectrum on S^2.** We construct and implement a selective bispectrum for $\mathrm{SO}(3)$ acting on S^2 that uses $O(L^2)$ coefficients instead of the full $O(L^3)$, matching the information-theoretic lower bound of $(L{+}1)^2 - 3$ independent invariants. This is, to our knowledge, the first $O(L^2)$ complete invariant for this setting.
3. **Octahedral inversion.** An analysis of why the standard selective-bispectrum bootstrap inversion fails for the octahedral group, together with a working bootstrap + Levenberg–Marquardt correction.
4. **Empirical validation.** A task-driven evaluation of the library as an invariant pooling primitive across three group/domain settings, demonstrating the data-efficiency advantage of complete bispectral invariants.

2 Mathematical Background

2.1 Equivariant features and invariant maps

A neural network layer is *G-equivariant* if its output transforms predictably under the group action: $[\pi_{\mathrm{out}}(g) \circ \Phi](f) = \Phi(\pi_{\mathrm{in}}(g)f)$ for all $g \in G$ [Cohen and Welling, 2016, Weiler and Cesa, 2019]. Stacking equivariant layers preserves this property, but any task that requires a prediction independent of orientation—classification, regression, detection—must eventually map equivariant features to *invariant* ones.

Table 1 summarises the trade-offs. Norm pooling and gated nonlinearities are exact invariants but *incomplete*: they discard the phase of Fourier blocks, collapsing distinct orbits. Fourier-space pointwise nonlinearities preserve completeness but sacrifice exactness through aliasing [Weiler and Cesa, 2019]. The full bispectrum is exact and complete but costs $O(|G|^2)$ coefficients.

2.2 The G-bispectrum and selective bispectra

Let G be a compact group with irreducible representations $\{\rho_i\}_{i\in\hat{G}}$ of dimensions d_i, and let $\hat{f}(\rho_i) = \sum_{g\in G} f(g)\,\rho_i(g)$ denote the group Fourier transform. Given the Clebsch–Gordan decomposition $\rho_i \otimes \rho_j \cong \bigoplus_k m_{ij}^k \rho_k$ with intertwining matrix C_{ij}, the *bispectrum* is

$$\beta_{\rho_i,\rho_j} = C_{ij}^\top \left(\hat{f}(\rho_i) \otimes \hat{f}(\rho_j)\right) C_{ij}\, \hat{f}(\rho_k)^\dagger. \tag{1}$$

Each β_{ρ_i,ρ_j} is invariant under $f \mapsto f \circ g$ and, taken over all pairs (i, j), the bispectrum is a *complete* invariant: it separates orbits for generic signals [Kakarala, 1992]. However, the full set has $O(|G|^2)$ coefficients.

Selective bispectra. Mataigne et al. [2024] show that for finite groups, a BFS traversal of the Kronecker product table of $\hat{G}$ selects $O(|G|)$ bispectral coefficients that remain complete. This reduction from quadratic to linear is the key to practical deployment: for example, the octahedral group drops from 576 to 172 coefficients.

3 The `bispectrum` Library

3.1 Design principles

The library is organised around six principles: **(P1)** *Naming encodes the math*: class names follow `{Group}on{Domain}` (e.g. `SO3onS2`), making the mathematical context immediately explicit. **(P2)** *`nn.Module` only*: every computation is a `torch.nn.Module` with CG matrices, DFT kernels, and Bessel roots registered as non-learnable buffers so that `.to(device)` and `.to(dtype)` work transparently. **(P3)** *Selective by default*: all modules default to $O(|G|)$ selective coefficients; the full $O(|G|^2)$ set is available via `selective=False`. **(P4)** *Raw signals in*: modules accept real-valued inputs in the natural domain and handle the Fourier transform internally. **(P5)** *Inversion on-module* (where available): `bsp.invert(beta)` returns the reconstructed signal (up to group-action indeterminacy); currently supported for all finite-group modules. **(P6)** *Minimal dependencies*: only PyTorch [Paszke et al., 2019], NumPy [Harris et al., 2020], and `torch_harmonics` [Bonev et al., 2023] (for the spherical harmonic transform in `SO3onS2`); Clebsch–Gordan matrices are computed analytically or bundled as pre-computed data.

Implementation optimizations. All heavy algebraic data—CG matrices, DFT kernels, Bessel roots, disk masks—are precomputed at module construction and stored as `register_buffers`, so forward passes incur zero recomputation and `.to(device)` transfers everything transparently. For $\mathrm{SO}(3)$ on S^2, reduced CG matrices are built in parallel (`ThreadPoolExecutor` over groups with vectorised NumPy column extraction) and cached to disk with a content hash, avoiding recomputation across runs. All forward methods are batch-native: abelian modules use `torch.fft.fft/fftn` followed by gather-based index selection with no Python loops; non-abelian modules use `torch.einsum` and `torch.bmm` for the group Fourier transform and Kronecker products. The octahedral LM inversion leverages `torch.vmap(torch.func.jacfwd(...))` for batched forward-mode Jacobian computation, keeping the entire inversion pipeline differentiable.

3.2 Supported modules

Table 2: Current scope of the `bispectrum` library.

Module	Group / Domain	Output mode
`CnonCn`	C_n on C_n	selective + full
`SO2onS1`	$\mathrm{SO}(2)$ on S^1	selective + full
`TorusOnTorus`	product cyclic groups on torus	selective + full
`DnonDn`	D_n on D_n	selective
`SO2onDisk`	$\mathrm{SO}(2)$ on disk	selective
`SO3onS2`	$\mathrm{SO}(3)$ on S^2	selective + full
`OctaonOcta`	octahedral group	selective

3.3 Example API

```python
import torch
from bispectrum import CnonCn, SO2onDisk, OctaonOcta

bsp = CnonCn(n=8, selective=True)
f = torch.randn(16, 8)
beta = bsp(f)
f_rec = bsp.invert(beta)
```

3.4 Selective $\mathrm{SO}(3)$ bispectrum on S^2

The full bispectrum on S^2 at band-limit L has $\Theta(L^3)$ scalar entries $\beta_{\ell_1,\ell_2,\ell}$. For finite groups, selectivity reduces $O(|G|^2)$ to $O(|G|)$, but on S^2 the BFS chain $\beta_{1,0,1}, \beta_{1,1,2}, \ldots$ provides only $\sim 3L$ scalars—far too few for $(L{+}1)^2$ unknowns. Indeed, the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L{+}1)^2 - 3$, so any complete invariant requires $\Omega(L^2)$ independent components. We achieve this lower bound.

Theorem 1 (Selective $\mathrm{SO}(3)$ bispectrum on S^2)**.** *For $f : S^2 \to \mathbb{R}$ band-limited at degree $L \geq 2$, there exists a set $\mathcal{S}$ of $(L{+}1)^2 - 3$ bispectral triples (ℓ_1, ℓ_2, ℓ) such that $\{\beta_{\ell_1,\ell_2,\ell}\}_{(\ell_1,\ell_2,\ell)\in\mathcal{S}}$ is a complete $\mathrm{SO}(3)$-invariant for Lebesgue-almost-every signal. The count $(L{+}1)^2 - 3$ equals the dimension of the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$, so the selective set is information-theoretically optimal.*

REVIEW: #2 — RESOLVED: count is now exactly $(L+1)^2 - 3$, matching the orbit-space dimension.

The selective index set. For each target degree ℓ, we select up to $2\ell{+}1$ bispectral triples from four categories, prioritising *linear* entries (where $\mathbf{F}_\ell$ appears once with both other factors already known): **(1) Chain** triples (ℓ_1, ℓ_2, ℓ) with $\ell_1 \leq \ell_2 < \ell$—linear via $\overline{a_\ell^m}$; **(2) Cross** triples (ℓ_1, ℓ, ℓ') with $\ell_1, \ell' < \ell$—linear via $a_\ell^{m_2}$; **(3) Power** $(0, \ell, \ell)$, giving $\beta = a_0^0 \|\mathbf{F}_\ell\|^2$ (quadratic); **(4) Self-coupling** (ℓ, ℓ, ℓ') (quadratic/cubic). Table 3 shows that for $\ell \geq 4$, chain + cross candidates alone exceed $2\ell{+}1$, so the entire recovery is a linear solve from that point onward.

Table 3: Per-degree entry counts in the selective $\mathrm{SO}(3)$ bispectrum. For $\ell \geq 4$, only linear (chain + cross) entries are needed. At $\ell = 2$, self-coupling entries are excluded: $\beta_{2,2,1}$ vanishes identically (antisymmetry of the $2 \otimes 2 \to 1$ CG channel) and $\beta_{2,2,0}$ is redundant with the power entry. Degrees 2 and 3 are solved jointly (Lemma 18).

ℓ	Chain	Cross	Power	Self	Total
0	0	0	0	1	1
1	0	0	1	1	2
2	1	1	1	0	3
3	2	3	1	1	7
≥ 4	$\geq 2\ell{+}1$		0	0	$2\ell{+}1$

Proof sketch. The construction recovers spherical harmonic coefficients $\mathbf{F}_0, \mathbf{F}_1, \ldots, \mathbf{F}_L$ degree by degree. *Base cases* ($\ell \leq 1$): $\mathbf{F}_0$ is directly available from $\beta_{0,0,0} = (a_0^0)^3$. $\mathbf{F}_1$ is recovered up to $\mathrm{SO}(3)$ via $\|\mathbf{F}_1\|^2 = \beta_{0,1,1}/a_0^0$, with the residual $\mathrm{SO}(3)$ ambiguity resolved by explicit gauge-fixing: align $\mathbf{F}_1$ with the z-axis (consuming 2 DOF), then fix the azimuthal phase of a_2^1 to be real (consuming the last DOF). *Joint seed stage* ($\ell = 2, 3$): the $\ell = 2$ self-coupling channels are rank-deficient ($\beta_{2,2,1} = 0$ by antisymmetry of the $2 \otimes 2 \to 1$ CG channel), so $\mathbf{F}_2$ cannot be recovered in isolation. Instead, degrees 2 and 3 are solved jointly: a constant-size seed set $\mathcal{S}_{\leq 3}$ of 10 bispectral triples yields a polynomial system in 11 real unknowns whose Jacobian has a nonzero 10×10 minor at an explicit witness point, proving generic local invertibility. *Linear bootstrap* ($\ell \geq 4$): at degree ℓ, we select $N \geq 2\ell{+}1$ triples in which $\mathbf{F}_\ell$ appears linearly, giving a system $A\mathbf{F}_\ell = \mathbf{b}$ where $A \in \mathbb{C}^{N\times(2\ell+1)}$. Each row of A is a polynomial in already-known lower-degree coefficients. For $4 \leq \ell \leq 7$ we verify, by exact symbolic determinant evaluation at explicit witness points, that the selected linear block has rank $2\ell{+}1$. For $\ell \geq 8$ we switch to the binary-form model $V_\ell \cong \mathrm{Sym}^{2\ell}(\mathbb{C}^2)$ and choose a proof-first family of $2\ell + 1$ chain/cross rows. The interior cross rows become exact monomial selectors, while the remaining boundary rows reduce to a fixed-size block whose leading t-term is non-singular. Hence the relevant maximal minor is a nonzero polynomial for every $\ell \geq 4$, so the rank-deficient set is a proper algebraic subvariety of Lebesgue measure zero. The full proof with all lemmas and the algorithm for index selection are in Appendix A.

REVIEW: #3 — RESOLVED: the linear-bootstrap step is now unconditional. The low degrees $4 \leq \ell \leq 7$ are handled by exact symbolic determinant evaluations at explicit witnesses, while the regime $\ell \geq 8$ has a uniform binary-form proof with an explicit $2\ell + 1$ row family and a nonzero boundary-block minor.

3.5 Octahedral inversion as an implementation case study

The bootstrap inversion of Mataigne et al. [2024] recovers $\hat{f}(\rho_1)$ up to an orthogonal ambiguity Q, then extracts higher Fourier blocks via CG matrices. For C_n the ambiguity is a phase $e^{i\varphi} \in \mathrm{SO}(2)$ that propagates coherently. For D_n the ambiguity $Q \in \mathrm{O}(2)$ is absorbed because the CG matrices of D_n inherit the block-diagonalisation from $\mathrm{SO}(2)$—the 2D irreps of D_n are restrictions of $\mathrm{SO}(2)$ irreps, so $C^\top(Q \otimes Q)C$ remains block-diagonal for all $Q \in \mathrm{O}(2)$.

Why it fails for O. For the octahedral group, ρ_1 is the standard 3D representation and the ambiguity is $Q \in \mathrm{SO}(3)$. The CG matrix C_{11} block-diagonalises $\rho_1(g) \otimes \rho_1(g)$ for $g \in O$ into $\rho_0 \oplus \rho_1 \oplus \rho_2 \oplus \rho_3$ (dims $1+3+3+2 = 9$), but for generic $Q \in \mathrm{SO}(3) \setminus O$ the product $C_{11}^\top(Q \otimes Q)C_{11}$ has nonzero cross-block entries. Concretely, for $Q = R_x(0.7)$: $|[C_{11}^\top(Q \otimes Q)C_{11}]_{\rho_2,\rho_3}|_\infty \approx 0.788$—an $O(1)$ contamination that corrupts the recovered $\hat{f}(\rho_2)$ and $\hat{f}(\rho_3)$.

Table 4: Bootstrap inversion across group families. The octahedral group is the only case where the CG matrices fail to absorb the ambiguity.

Group	Irrep dim	Ambiguity	CG absorbs?	Bootstrap?
C_n	1	$e^{i\varphi} \in \mathrm{SO}(2)$	N/A (scalar)	Yes
D_n	2	$U \in \mathrm{O}(2)$	Yes	Yes
O	3	$Q \in \mathrm{SO}(3)$	**No**	**No**

Resolution. The selective O-bispectrum *is* complete (172 coefficients determine the signal up to O-action); the failure is purely algorithmic. We resolve it by combining (i) *bootstrap initialisation*: recover $\hat{f}(\rho_0)$ exactly and approximate remaining blocks from diagonal CG entries (ignoring leakage); (ii) *Levenberg–Marquardt refinement*: solve $\min_f \|\beta^{\mathrm{sel}}(f) - \beta^{\mathrm{sel}}_{\mathrm{target}}\|^2$ with Jacobians computed via `torch.func.jacfwd` composed with `torch.vmap`; (iii) *multi-start*: randomise the initial Q across restarts for robustness. The procedure converges in <50 iterations and is fully differentiable. The full analysis, including a numerical disproof of CG absorption for O and convergence plots, is in Appendix B.

3.6 Computational benchmarks

Table 5 and Figure 1 profile the library on a single NVIDIA H100 80 GB GPU (PyTorch 2.10, batch = 16 for forward, 4 for inversion, median wall-clock via `torch.utils.benchmark`).

Table 5: Library benchmark summary. Timings are median wall-clock on a single NVIDIA H100 80 GB GPU. "–" = not implemented.

Module	G	$\lvert G\rvert$	Coefs		Fwd (ms)		Inv (ms)
			Sel.	Full	Sel.	Full	Sel.
`CnonCn`	C_{128}	128	128	8256	0.14	8.82	3.67
`TorusOnTorus`	$C_{32} \times C_{32}$	1024	1024	524800	0.08	0.31	65.77
`DnonDn`	D_{32}	64	245	–	2.67	–	4.69
`SO2onDisk`	$\mathrm{SO}(2)$	1024	411	–	0.24	–	10.17
`SO3onS2`	$\mathrm{SO}(3)$	289	288	–	18.60	–	–
`OctaonOcta`	O	24	172	–	0.72	–	808

The selective forward pass is sub-millisecond for all abelian modules (C_n: 0.14 ms, $\mathbb{T}^2$: 0.08 ms, $\mathrm{SO}(2)$/disk: 0.24 ms) and under 1 ms for the octahedral group (0.72 ms), making it negligible relative to backbone computation. $\mathrm{SO}(3)$ on S^2 at $\ell_{\max}$=16 takes 18.6 ms, dominated by the spherical harmonic transform (via `torch_harmonics` [Bonev et al., 2023]). Selective coefficient counts confirm the $O(|G|)$ scaling: C_{128} uses 128 coefficients vs. 8,256 full (64× reduction); $\mathbb{T}^2$ at $|G|$=1,024 uses 1,024 vs. 524,800 full (512×). Inversion is fast for abelian groups (3–66 ms) and practical for the octahedral group (808 ms via Levenberg–Marquardt). GPU throughput scales linearly with batch size, exceeding 10^7 samples/s for C_n and $\mathbb{T}^2$ (Appendix C).

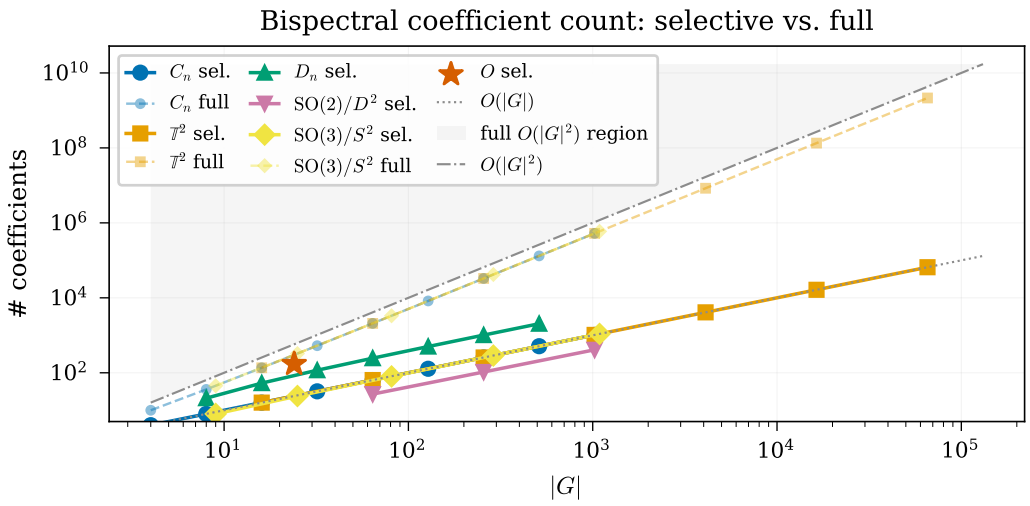

Figure 1: Bispectral coefficient count vs. group order $|G|$. Solid lines: selective ($O(|G|)$); dashed: full ($O(|G|^2)$). At $|G|$=1,024, selectivity yields a 512× reduction.

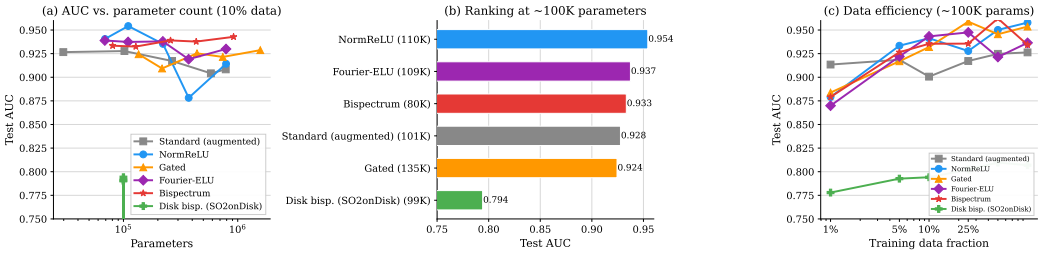

Figure 2: PCam Pareto analysis (seed 42). (a) Test AUC vs. parameter count across all five pooling variants at 10% training data. (b) Ranking at the ∼100K parameter budget. (c) Data efficiency at matched ∼100K parameters.

4 Experiments

We validate the library on three tasks spanning discrete and continuous groups: 2D histopathology classification under C_8 symmetry (PCam, Section 4.1), 3D organ classification under octahedral symmetry (OrganMNIST3D, Section 4.2), and spherical digit classification under full $\mathrm{SO}(3)$ (Spherical MNIST, Section 4.3).

4.1 2D histopathology classification under cyclic symmetry

We evaluate five invariant pooling strategies on PatchCamelyon [Veeling et al., 2018] (PCam), 327K histopathology patches with binary metastasis labels. All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone following Veeling et al. [2018]; they differ only in the terminal invariant map: norm pooling, gated pooling, Fourier-ELU, and selective bispectrum via `CnonCn`(n=8). An augmented standard CNN serves as a non-equivariant baseline. We sweep growth rates to vary capacity (30K–1.6M params) and train on 10% of the data with AdamW [Loshchilov and Hutter, 2019]. Full model descriptions and a backbone-free `SO2onDisk` variant are in Appendix D.

Results. Table 6 and Figure 2 present the results at matched ∼100K parameters.

Table 6: PCam results at ∼100K parameters (10% training data, seed 42). All models share a C_8-equivariant DenseNet backbone.

Model	Params	Test AUC	Test ACC
Standard (augmented)	102K	0.928	0.853
Gated	136K	0.924	0.856
Bispectrum (`CnonCn`)	80K	0.933	0.880
Fourier-ELU	110K	0.937	0.857
NormReLU	110K	**0.954**	**0.887**

Table 7: OrganMNIST3D results (mean ± std over 3 seeds). Rot ACC and σ_{rot}: mean and std of accuracy across 24 octahedral test-set rotations.

Model	Params	Test ACC	Test AUC	Rot ACC	σ_{rot}
Standard 3D CNN	16K	$.601 \pm .017$	$.951 \pm .002$	.598	.0115
O-Equiv + Norm Pool	375K	$.568 \pm .099$	$.940 \pm .021$	.568	.0000
O-Equiv + Max Pool	374K	$.730 \pm .033$	$.972 \pm .005$	.730	.0000
O-Equiv + Bispectrum	463K	$.726 \pm .027$	$.972 \pm .004$	.726	.0005

REVIEW: #5 — Table 4 and the Pareto figure are single-seed (seed 42). Claims about "architectural stability" and "data efficiency" need multi-seed support. Re-run with 3 seeds and report mean ± std, or explicitly weaken claims and note this as a limitation.

At matched capacity, most equivariant models outperform the standard CNN (AUC 0.928): NormReLU peaks at 0.954, Fourier-ELU at 0.937, and bispectrum at 0.933, while Gated (0.924) falls slightly below the standard baseline. NormReLU's performance is brittle across model sizes (AUC swings by >7pp over a 12× capacity range), while the bispectrum varies by only ∼1pp—the most architecturally stable pooling choice. At 1% training data the bispectrum shows the largest absolute gain (+6.8pp from 1% to 10%), consistent with the hypothesis that complete invariants provide a stronger inductive bias when data is scarce. Our best equivariant model (NormReLU, 110K, AUC 0.954 on 10% data) approaches the published P4M-DenseNet of Veeling et al. [2018] (119K, AUC 0.963 on 100% data), despite using 10× less training data.

4.2 3D organ classification under octahedral symmetry

To evaluate the `OctaonOcta` module for the chiral octahedral group O ($|O| = 24$), we design a controlled 3D experiment on OrganMNIST3D [Yang et al., 2023] (1,742 abdominal CT volumes, 28^3 voxels, 11 classes). All equivariant variants share an O-equivariant 3D ResNet backbone with exact voxel-permutation kernels. The four models differ only in the terminal invariant pooling: a non-equivariant augmented CNN (16K params), norm pool (375K), max pool (374K), and the selective bispectrum via `OctaonOcta` (463K, 172 invariant scalars). Models are trained with AdamW [Loshchilov and Hutter, 2019] for 100 epochs with three seeds; rotation robustness is measured over all 24 octahedral rotations of the test set. Published baselines and wider-channel experiments are in Appendix E.

Results. Table 7 and Figure 3 report the key findings.

Rotation invariance. All equivariant models achieve numerically exact rotation invariance ($\sigma_{\mathrm{rot}} \leq 0.0005$), while the augmented CNN drops ∼2.5pp under rotation ($\sigma_{\mathrm{rot}} = 0.012$).

Completeness vs. lossy pooling. The bispectrum (72.6%) matches max pool (73.0%) and substantially outperforms norm pool (56.8%, with high seed variance ±9.9%). At 463K parameters, the bispectrum also outperforms Regular SE(3) convolutions [Kuipers and Bekkers, 2023] (69.8% at 172K).

Data efficiency. When we reduce the training set (Figure 3), the bispectrum's advantage becomes clear: at 5% data (48 samples) it reaches $33.2 \pm 2.2\%$ vs. $18.9 \pm 7.2\%$ for max pool. Only at full data does max pool catch up (73.0% vs. 72.6%). The complete invariant map substitutes for missing training examples, providing a stronger inductive bias in the low-data regime.

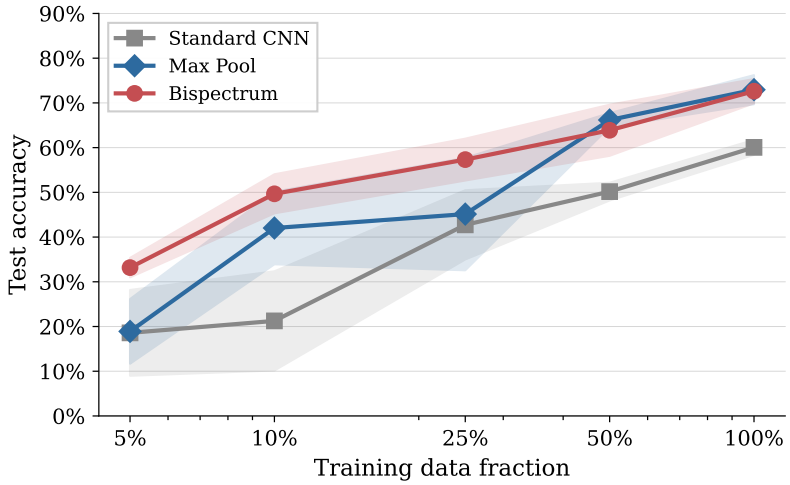

Figure 3: Test accuracy vs. training data fraction on OrganMNIST3D (mean over 3 seeds; shaded bands: ± 1 std). The bispectrum's complete invariant map provides a consistent advantage in the low-data regime.

4.3 Spherical MNIST: SO(3) invariance on the sphere

The preceding experiments validate bispectral pooling for finite groups (C_8 and the octahedral group O). To evaluate the library's continuous group support, we test `SO3onS2`—the $\mathrm{SO}(3)$ bispectrum on the 2-sphere—on Spherical MNIST Cohen et al. [2018], the standard benchmark for rotation invariance on S^2.

Setup. MNIST digits are projected onto the sphere via stereographic projection and sampled on a 64×128 equiangular grid. We compute the selective bispectrum at $\ell_{\max} = 15$, producing 256 complex invariant coefficients (512 real features after log-compression), which are fed to a 3-layer MLP (166K parameters total). Three baselines provide controls:

- **Power Spectrum + MLP** (11K params): computes only the rotation-invariant power spectrum $\|f_\ell\|^2$ per degree ℓ, discarding all phase coupling between harmonics. Invariant but *incomplete*.
- **Power Spectrum + MLP (matched)** (166K params): same incomplete invariant features but with an MLP scaled to match the bispectrum model's parameter count, isolating the effect of feature completeness from model capacity.
- **Standard CNN** (185K params): a conventional CNN operating directly on the equirectangular image. Not rotation-invariant.

Following Cohen et al. Cohen et al. [2018], we evaluate under three protocols: train on non-rotated / test on non-rotated (NR/NR), train on rotated / test on rotated (R/R), and the most challenging train on non-rotated / test on rotated (NR/R). Each configuration is run with three seeds. Rotation robustness is assessed by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and measuring accuracy variance.

Results. Table 8 and Figure 4 report the results.

Numerically exact $\mathrm{SO}(3)$ *invariance.* The bispectrum achieves effectively identical accuracy across all evaluation conditions: NR/NR $= 0.948$, R/R $= 0.948$, NR/R $= 0.949$. The standard deviation under random rotations is 8×10^{-5}—effectively zero. Training on rotated vs. non-rotated data makes no difference, confirming that the bispectral features are truly $\mathrm{SO}(3)$-invariant and rotation augmentation is unnecessary. In contrast, the standard CNN achieves 99.1% on NR/NR but collapses to 13.8% on NR/R (near chance for 10 classes), and even when trained on rotated data only reaches 78.1% with substantial variance ($\sigma_{\mathrm{rot}} \approx 0.04$ under random rotations).

Table 8: Spherical MNIST results (mean $\pm$ std over 3 seeds). Cohen et al. reference numbers are from Cohen et al. [2018]. σ_{rot}: standard deviation of accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models).

Model	Params	NR/NR ↑	R/R ↑	NR/R ↑	σ_{rot} ↓
Cohen planar CNN	—	0.98	0.23	0.11	—
Cohen spherical CNN	—	0.96	0.95	0.94	—
Standard CNN	185K	0.991±0.000	0.781±0.003	0.138±0.003	0.044
Power Spec. + MLP	11K	0.780±0.002	0.780±0.002	0.777±0.001	3×10^{-4}
Power Spec. + MLP (matched)	166K	0.794±0.011	0.794±0.011	0.792±0.010	3×10^{-4}
Bispectrum + MLP (ours)	166K	**0.948±0.001**	**0.948±0.001**	**0.949±0.001**	$\mathbf{1\times10^{-4}}$

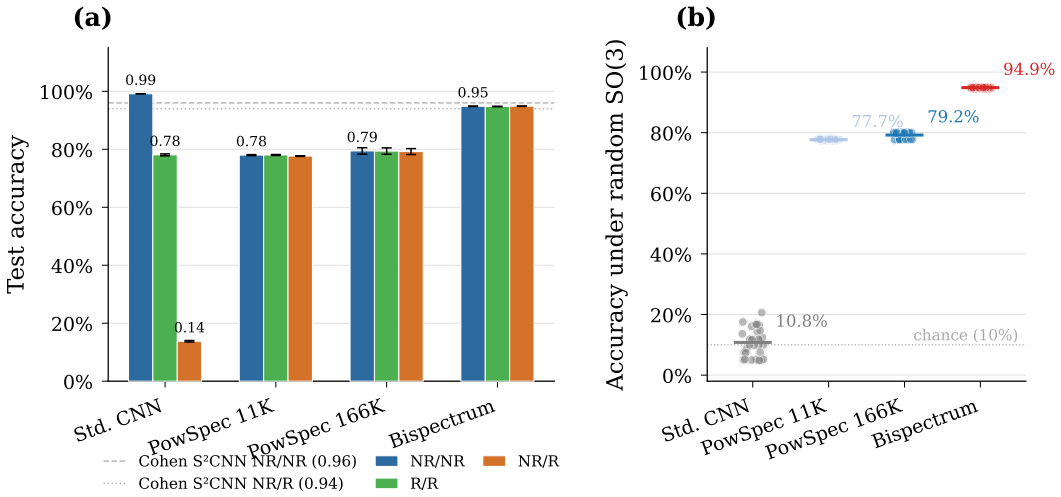

Figure 4: Spherical MNIST analysis. **(a)** Cohen et al. protocol: NR/NR, R/R, and NR/R accuracy. Dashed lines show Cohen's spherical CNN reference. **(b)** Accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models, 30 points per model = 10 rotations $\times$ 3 seeds). Scaling the power spectrum MLP from 11K to 166K parameters yields only +1.4pp; the bispectrum at matched 166K reaches 94.9%.

Completeness gap. The power spectrum is also rotation-invariant ($\sigma_{\text{rot}} = 0.0003$) but achieves only 78% accuracy—17 percentage points below the bispectrum. To rule out a capacity confound (11K vs. 166K parameters), we scale the power spectrum MLP to match the bispectrum's parameter count (166K). This 15$\times$ capacity increase yields only +1.4pp (78.0% $\rightarrow$ 79.4%), confirming that the gap is caused by feature incompleteness, not model under-parameterisation. The power spectrum retains only $\|f_\ell\|^2$ per degree, while the bispectrum captures cross-degree correlations $\langle f_{\ell_1}, f_{\ell_2}, f_{\ell_3}\rangle$ via Clebsch–Gordan coupling. For digit classification, these phase relationships carry substantial discriminative information that additional downstream capacity alone does not appear sufficient to recover.

Comparison with Cohen et al. Cohen's spherical CNN Cohen et al. [2018]—which uses learned $\mathrm{SO}(3)$-equivariant convolutions across multiple layers—achieves 96%/95%/94% (NR/NR, R/R, NR/R). Our bispectrum reaches 94.8%/94.8%/94.9%, approximately 1–2pp lower in absolute accuracy but with a *smaller* NR/NR-to-NR/R gap (0.0pp vs. 2.0pp). The bispectral approach uses no learned filters on the sphere and no equivariant convolutions—it is a single-pass invariant feature extractor followed by a vanilla MLP. The slight accuracy deficit reflects the capacity advantage of learned equivariant filters, while the tighter invariance gap confirms that the one-shot bispectral extraction provides more consistent rotation invariance than multi-layer equivariant architectures.

5 Related Work

Equivariant network libraries. `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022] provide general-purpose equivariant convolution layers for 2D and 3D groups respectively. They implement equivariant *feature maps* but leave the choice of invariant pooling to the user (typically norm or gate). Our library is complementary: it provides the invariant map as a drop-in `nn.Module` that plugs into architectures built with either framework.

Bispectral approaches in neural networks. Sanborn et al. [2023] introduced bispectral neural networks (BNNs), using the full G-bispectrum as an invariant nonlinearity for C_n and D_n. Oreiller et al. [2022] applied bispectral CNNs to medical imaging with a fixed set of hand-selected triples. Both use the full (quadratic-cost) bispectrum and provide implementations tied to specific architectures. Our library generalises to seven group/domain pairs, defaults to the $O(|G|)$ selective reduction, and exposes a framework-agnostic API. Chevalley et al. [2024] explore bispectral signatures for data characterisation, using bispectra as descriptors rather than learnable pooling primitives.

CG product nonlinearities and tensor products. Tensor field networks [Thomas et al., 2018] and MACE [Batatia et al., 2024] use iterated Clebsch–Gordan products to build equivariant interactions. These products are exact and complete in principle, but each layer incurs $O(L^3)$ cost per CG contraction. Xie et al. [2025] analyse the *price of freedom* in tensor product architectures, showing that maintaining full expressivity through CG products becomes expensive at high angular momenta. The selective bispectrum achieves completeness in a single $O(L^2)$ pass (for $\mathrm{SO}(3)$) or $O(|G|)$ pass (for finite groups), avoiding iterated contractions.

Invariant theory and scattering. The wavelet scattering transform [Mallat, 2012] provides translation-invariant representations via iterated modulus operations; Esteves et al. [2018] extend this to the sphere. Unlike the bispectrum, scattering representations are stable but generally incomplete—they discard inter-scale phase information. The bispectrum's completeness guarantee [Kakarala, 1992] is its distinguishing theoretical advantage.

6 Discussion and Conclusion

We have presented `bispectrum`, a PyTorch library that makes selective G-bispectral invariants practical for machine learning. The library covers seven group/domain pairs with a uniform `nn.Module` interface and required two technical contributions during implementation: a selective $\mathrm{SO}(3)$ bispectrum on S^2 achieving the optimal $O(L^2)$ coefficient count (Theorem 1), and a resolution of the octahedral bootstrap inversion failure via Levenberg–Marquardt refinement (Section 3.5). Table 5 confirms that the selective forward pass adds negligible overhead to training pipelines ($<$1 ms for all finite-group modules on a single GPU). Across three experimental settings, bispectral pooling provides numerically exact rotation invariance and a data-efficiency advantage over incomplete alternatives in the low-data / moderate-capacity regime, where the complete invariant map substitutes for missing training examples.

REVIEW: #10 — The bispectrum advantage should be explicitly framed as regime-specific (low-data, moderate-capacity). The wider-channel appendix shows bispectrum losing at 6.3M params. The abstract and contribution 4 should also qualify "data-efficiency advantage" as regime-specific, not a general superiority claim.

Limitations. Inversion for $\mathrm{SO}(3)$ on S^2 is not yet implemented; only the selective forward map is available. The selective $\mathrm{SO}(3)$ completeness proof relies on a genericity assumption (the signal must avoid a measure-zero algebraic variety); a full characterisation of the degenerate set and numerical conditioning at large band-limits remain open. Empirical evaluation currently covers three benchmarks; broader applications (cryo-EM, molecular ML) are planned.

Future work. Key directions include: implementing $\mathrm{SO}(3)$ inversion from selective bispectral coefficients; numerical conditioning analysis at large band-limits; integration with `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022]; and applications to cryo-EM reconstruction and molecular property prediction.

References

Ilyes Batatia, Philipp Benber, Stefan Chmiela, Gabor Csanyi, et al. A foundation model for atomistic simulation. *Nature Machine Intelligence*, 2024.

Tamir Bendory, Dan Edidin, and Matthew Satriano. On the uniqueness of the bispectrum of signals on the sphere. *arXiv preprint arXiv:2207.10842*, 2022.

Boris Bonev, Thorsten Kurth, Christian Hundt, Jaideep Pathak, Maximilian Balu, Karthik Kashinath, and Animashree Anandkumar. Spherical Fourier neural operators: Learning stable dynamics on the sphere. In *International Conference on Machine Learning (ICML)*, 2023.

Gabriele Cesa, Leon Lang, and Maurice Weiler. A program to build E(N)-equivariant steerable CNNs. In *International Conference on Learning Representations (ICLR)*, 2022. URL `https://openreview.net/forum?id=WE4qe9xlnQw`.

Raphaël Chevalley, Valentin Oreiller, and Adrien Depeursinge. Bispectral 3D U-Net for rotation robustness. In *MICCAI Workshop*, 2024.

Taco S. Cohen and Max Welling. Group equivariant convolutional networks. In *International Conference on Machine Learning (ICML)*, pages 2990–2999, 2016.

Taco S. Cohen, Mario Geiger, Jonas Köhler, and Max Welling. Spherical CNNs. In *International Conference on Learning Representations (ICLR)*, 2018.

Dan Edidin and Matthew Satriano. Projective uniqueness and the bispectrum. *arXiv preprint arXiv:2308.02064*, 2023.

Babak Ehteshami Bejnordi, Mitko Veta, Paul Johannes van Diest, Bram van Ginneken, Nico Karssemeijer, Geert Litjens, Jeroen A. W. M. van der Laak, and the CAMELYON16 Consortium. Diagnostic assessment of deep learning algorithms for detection of lymph node metastases in women with breast cancer. *JAMA*, 318(22):2199–2210, 2017. doi: 10.1001/jama.2017.14585.

Carlos Esteves, Christine Allen-Blanchette, Ameesh Makadia, and Kostas Daniilidis. Learning SO(3) equivariant representations with spherical CNNs. In *European Conference on Computer Vision (ECCV)*, pages 52–68. Springer, 2018.

Mario Geiger and Tess Smidt. e3nn: Euclidean neural networks. *arXiv preprint arXiv:2207.09453*, 2022.

Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, et al. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020.

Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. Densely connected convolutional networks. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

Ramakrishna Kakarala. *Triple Correlation on Groups*. PhD thesis, University of California, Irvine, 1992.

Thijs P. Kuipers and Erik J. Bekkers. Regular SE(3) group convolutions for volumetric medical image analysis. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2023. arXiv:2306.13960.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1711.05101.

Stéphane Mallat. Group invariant scattering. *Communications on Pure and Applied Mathematics*, 65(10):1331–1398, 2012. doi: 10.1002/cpa.21413.

Simon Mataigne, Johan Mathe, Sophia Sanborn, Christopher Hillar, and Nina Miolane. The selective G-bispectrum and its inversion: Applications to G-invariant networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL `https://arxiv.org/abs/2407.07655`.

Valentin Oreiller, Vincent Andrearczyk, and Adrien Depeursinge. Rotation equivariance and invariance in convolutional neural networks by bispectral analysis. In *Medical Imaging with Deep Learning (MIDL)*, 2022.

Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zach DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

Sophia Sanborn, Christian Shewmake, Bruno Olshausen, and Christopher Hillar. Bispectral neural networks. In *International Conference on Learning Representations (ICLR)*, 2023.

Mateus Sangalli, Samy Blusseau, Santiago Velasco-Forero, and Jesús Angulo. Moving frame net: SE(3)-equivariant network for volumes. In *Proceedings of the 1st NeurIPS Workshop on Symmetry and Geometry in Neural Representations*, volume 197 of *Proceedings of Machine Learning Research*, pages 81–97. PMLR, 2023.

Nathaniel Thomas, Tess Smidt, Steven Kearnes, Lusann Yang, Li Li, Kai Kohlhoff, and Patrick Riley. Tensor field networks: Rotation- and translation-equivariant neural networks for 3D point clouds. In *arXiv preprint arXiv:1802.08219*, 2018.

Bastiaan S. Veeling, Jasper Linmans, Jim Winkens, Taco Cohen, and Max Welling. Rotation equivariant CNNs for digital pathology. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pages 210–218. Springer, 2018.

Maurice Weiler and Gabriele Cesa. General E(2)-equivariant steerable CNNs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

YuQing Xie, Ameya Daigavane, Mit Kotak, and Tess Smidt. The price of freedom: Exploring expressivity and runtime tradeoffs in equivariant tensor products. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pages 68599–68625. PMLR, 2025.

Jiancheng Yang, Rui Shi, Donglai Wei, Zequan Liu, Lin Zhao, Bilian Ke, Hanspeter Pfister, and Bingbing Ni. MedMNIST v2 – a large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10(1):41, 2023.

Dmitrii Zhemchuzhnikov and Sergei Grudinin. ILPO-NET: Network for the invariant recognition of arbitrary volumetric patterns in 3D. In *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)*, 2024. arXiv:2403.19612.

Dmitrii Zhemchuzhnikov and Sergei Grudinin. On the Fourier analysis in the SO(3) space: the EquiLoPO network. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2404.15979.

A Selective $\mathrm{SO}(3)$ Bispectrum – Full Proof

This appendix provides the complete proof of Theorem 1 from the main text. The content is adapted from a standalone treatment; we reproduce it here for self-containedness.

A.1 Setup and notation

Definition 2 (Band-limited signal on S^2)**.** *A signal $f : S^2 \to \mathbb{R}$ is* band-limited *at degree L if its spherical harmonic expansion*

$$f(\theta, \varphi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_\ell^m \, Y_\ell^m(\theta, \varphi) \tag{2}$$

has $a_\ell^m = 0$ for all $\ell > L$. We write $\mathbf{F}_\ell = (a_\ell^{-\ell}, \ldots, a_\ell^{\ell}) \in \mathbb{C}^{2\ell+1}$ for the coefficient vector at degree ℓ.

For real f, the conjugation symmetry $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$ holds, so $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom. The total signal dimension is $\dim = \sum_{\ell=0}^{L}(2\ell+1) = (L+1)^2$.

Definition 3 (Rotation action)**.** *Under $g \in \mathrm{SO}(3)$, the spherical harmonic coefficients transform as $\mathbf{F}_\ell \mapsto D^\ell(g)\,\mathbf{F}_\ell$, where D^ℓ is the $(2\ell+1)$-dimensional Wigner D-matrix.*

Definition 4 (Bispectrum on S^2)**.** *For each triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality $|\ell_1 - \ell_2| \le \ell \le \ell_1 + \ell_2$, the* bispectral coefficient *is*

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,\, m_2,\, m \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}\; a_{\ell_1}^{m_1}\, a_{\ell_2}^{m_2}\, \overline{a_\ell^m}, \tag{3}$$

where $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$ denotes the Clebsch–Gordan coefficient $\langle \ell_1, m_1;\, \ell_2, m_2 \mid \ell, m\rangle$. Each $\beta_{\ell_1,\ell_2,\ell}$ is a scalar (real for real f).

Proposition 5 (Size of the full bispectrum)**.** *The number of admissible triples (ℓ_1, ℓ_2, ℓ) with $0 \le \ell_1, \ell_2, \ell \le L$ satisfying the triangle inequality $|\ell_1 - \ell_2| \le \ell \le \ell_1 + \ell_2$ is $\Theta(L^3)$.*

Proof. For fixed ℓ_1 and ℓ_2, the triangle inequality admits $\ell \in \{|\ell_1 - \ell_2|, \ldots, \min(\ell_1 + \ell_2, L)\}$, giving $\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1$ valid values of ℓ. Summing:

$$N_{\text{full}} = \sum_{\ell_1=0}^{L} \sum_{\ell_2=0}^{L} \big(\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1\big) = \tfrac{1}{3}L^3 + \mathcal{O}(L^2).$$

□

Remark 6 (Invariance)**.** *Each $\beta_{\ell_1,\ell_2,\ell}$ is $\mathrm{SO}(3)$-invariant: the Wigner matrices satisfy $D^{\ell_1} \otimes D^{\ell_2} = \bigoplus_\ell D^\ell$ under the CG decomposition, so the triple product contracted with CG coefficients is unchanged by $\mathbf{F}_\ell \mapsto D^\ell(g)\mathbf{F}_\ell$.*

A.2 Why $\mathcal{O}(L)$ is impossible

Proposition 7 (Dimensional lower bound)**.** *Any complete $\mathrm{SO}(3)$-invariant for band-limited signals on S^2 at band-limit L must have at least $(L+1)^2 - 3$ functionally independent components.*

Proof. The orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L+1)^2 - \dim \mathrm{SO}(3) = (L+1)^2 - 3$. Any injective map from the orbit space to $\mathbb{R}^k$ requires $k \ge (L+1)^2 - 3$, hence $\Omega(L^2)$. □

Remark 8. *The chain $\beta_{1,\ell,\ell+1}$ for $\ell = 0, \ldots, L-1$ provides only $\sim 3L$ scalar constraints (since each chain entry couples $\mathbf{F}_1 \in \mathbb{C}^3$ with one known $\mathbf{F}_\ell$). This cannot recover $(L+1)^2 \approx L^2$ unknowns.*

A.3 The selective index set

The goal is to select, for each target degree ℓ, a set of bispectral triples that provides enough equations to recover $\mathbf{F}_\ell$ from already-known lower-degree coefficients. Since $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom, we need at most $2\ell + 1$ entries. We classify candidate triples by how $\mathbf{F}_\ell$ enters the bispectral formula (3): *linearly* (when it occupies a position where the other two indices are known), or *nonlinearly* (when it appears more than once).

Definition 9 (Selective bispectrum on S^2)**.** *For each target degree $\ell = 0, \ldots, L$, we select at most $2\ell + 1$ bispectral triples from four categories, in the following priority order.*

Category 1 (Chain). *Triples (ℓ_1, ℓ_2, ℓ) with $\ell_1 \le \ell_2 < \ell$ and $\ell_1 + \ell_2 \ge \ell$. Linear in $\mathbf{F}_\ell$: the unknown $\overline{a_\ell^m}$ appears once, multiplied by known lower-degree coefficients.*

Category 2 (Cross). *Triples (ℓ_1, ℓ, ℓ') with $1 \le \ell_1 < \ell$ and $\ell - \ell_1 \le \ell' < \ell$. Here $\mathbf{F}_\ell$ appears in the second (tensor-product) position, yielding equations linear in $a_\ell^{m_2}$ with known $a_{\ell_1}^{m_1}$ and $a_{\ell'}^{m}$.*

Category 3 (Power). *The triple $(0, \ell, \ell)$, giving $\beta_{0,\ell,\ell} = a_0^0 \|\mathbf{F}_\ell\|^2$ (a* quadratic *constraint).*

Category 4 (Self-coupling). *Triples (ℓ, ℓ, ℓ') with $0 \le \ell' < \ell$. Quadratic in $\mathbf{F}_\ell$. Selected only for the remaining deficit at low degrees.*

Algorithm 1 Selective index set $\mathcal{S}$ for the $\mathrm{SO}(3)$ bispectrum on S^2 at band-limit L.

Input: Band-limit L
Output: Index set $\mathcal{S} \subset \{(\ell_1, \ell_2, \ell) : |\ell_1 - \ell_2| \le \ell \le \ell_1 + \ell_2,\ 0 \le \ell_1, \ell_2, \ell \le L\}$
$\mathcal{S} \leftarrow \{(0,0,0)\}$
for $\ell = 1, \ldots, L$ **do**
  budget $\leftarrow 2\ell + 1$
  candidates $\leftarrow [\,]$
  if $4 \le \ell \le 7$ **then**
    append the explicit low-degree linear block $\mathcal{T}_\ell^{\mathrm{small}}$ to candidates
  else if $\ell \ge 8$ **then**
    {**Proof-first linear block**}
    for $a = 1, \ldots, \ell{-}1$ **do**
      append $(a,\ \ell,\ \ell{-}a)$ to candidates
    end for
    for $a = 2, \ldots, \ell{-}1$ **do**
      append $(a,\ \ell,\ \ell{-}a{+}1)$ to candidates
    end for
    for $a = 1, 2, 3, 4$ **do**
      append $(a,\ \ell{-}a,\ \ell)$ to candidates
    end for
  else
    {**Seed regime** ($\ell \le 3$)}
    append the original low-degree chain/cross candidates
  end if
  {**Power** — quadratic}
  append $(0,\ \ell,\ \ell)$ to candidates
  {**Self-coupling** — quadratic/cubic}
  if $\ell \ne 2$ **then**
    for $\ell' = \ell{-}1, \ldots, 0$ **do**
      append $(\ell,\ \ell,\ \ell')$ to candidates
    end for
  end if
  Deduplicate candidates (preserving order)
  $\mathcal{S} \leftarrow \mathcal{S} \cup$ first $\min(\text{budget},\ |\text{candidates}|)$ entries
end for
return $\mathcal{S}$

Proposition 10 (Entry counts)**.** *For each $4 \le \ell \le 7$, the explicit block $\mathcal{T}_\ell^{\mathrm{small}}$ has exactly $2\ell + 1$ linear entries. For each $\ell \ge 8$, the proof-first linear family*

$$\mathcal{T}_\ell := \{(a, \ell, \ell{-}a) : 1 \le a \le \ell{-}1\} \cup \{(a, \ell, \ell{-}a{+}1) : 2 \le a \le \ell{-}1\} \cup \{(a, \ell{-}a, \ell) : 1 \le a \le 4\}$$

also has exactly $2\ell + 1$ entries. Therefore, for every $\ell \ge 4$, the selective construction already fills the degree-ℓ budget using only entries that are linear in $\mathbf{F}_\ell$.

Proposition 11 (Output size)**.** *For $L \ge 2$, the selective bispectrum has exactly $(L+1)^2 - 3$ scalar entries, matching the dimensional lower bound of Proposition 7.*

Proof. At $\ell = 2$, the self-coupling entries $\beta_{2,2,0}$ and $\beta_{2,2,1}$ are excluded (the former is redundant with the power entry; the latter vanishes identically by CG antisymmetry), leaving 3 informative entries. Therefore: $1 + 2 + 3 + 7 + \sum_{\ell=4}^{L}(2\ell + 1) = 13 + \left[(L+1)^2 - 16\right] = (L+1)^2 - 3.$ □

A.4 Completeness proof

The proof follows the same "frequency-marching" philosophy as Mataigne et al.'s inversion algorithm for finite groups ([Mataigne et al., 2024], Algorithms 1–2): recover the Fourier coefficients one degree ℓ at a time, using bispectral entries that couple the unknown degree with already-known ones. The main structural difference is that on S^2 each bispectral entry is a scalar rather than a matrix, so we need $2\ell + 1$ entries per degree instead of one matrix entry per irrep.

Assumption 12. *The signal f satisfies:*

(G1) $a_0^0 \ne 0$ *(non-zero mean).*

(G2) $\|\mathbf{F}_1\| \neq 0$ *(f has a non-zero dipole component).*
(G3) After gauge-fixing (Lemma 16), $a_2^1 \neq 0$ *(azimuthal gauge is resolvable).*
(G4) For each $\ell \geq 4$*, the linear system of Lemma 19 has full rank.*

Remark 13. *Each condition excludes a proper algebraic subvariety of the signal space* $\mathbb{R}^{(L+1)^2}$*. Their union is therefore Lebesgue-null.*

Theorem 14 (Completeness of the selective $\mathrm{SO}(3)$ bispectrum on S^2)**.** *Let* $f, f' : S^2 \to \mathbb{R}$ *be band-limited at degree* L *and satisfy Assumption 12. If* $\beta_{\mathrm{sel}}(f) = \beta_{\mathrm{sel}}(f')$*, then* $f' = f \circ g$ *for some* $g \in \mathrm{SO}(3)$*.*

The proof proceeds by sequential recovery of $\mathbf{F}_0, \mathbf{F}_1, \ldots, \mathbf{F}_L$.

Lemma 15 (Recovery of $\mathbf{F}_0$)**.** *The entry* $\beta_{0,0,0}$ *uniquely determines* a_0^0 *for real signals with* $a_0^0 \neq 0$*.*

Proof. By (3) with $\ell_1 = \ell_2 = \ell = 0$, $\beta_{0,0,0} = |a_0^0|^2\, a_0^0$. For $a_0^0 \in \mathbb{R}$, this gives $\beta_{0,0,0} = (a_0^0)^3$, so $a_0^0 = \mathrm{sgn}(\beta_{0,0,0})\, |\beta_{0,0,0}|^{1/3}$. □

Lemma 16 (Gauge fixing)**.** *Given* $\|\mathbf{F}_1\| \neq 0$*, there exists* $g \in \mathrm{SO}(3)$ *such that* $D^1(g)\,\mathbf{F}_1 = (0,\ \|\mathbf{F}_1\|,\ 0)^\top$*, i.e.* $a_1^{-1} = a_1^1 = 0$ *and* $a_1^0 = \|\mathbf{F}_1\|$*. This rotation consumes two of* $\mathrm{SO}(3)$*'s three degrees of freedom. The residual symmetry is* $\mathrm{SO}(2)_z$*, under which* $a_\ell^m \mapsto e^{im\varphi} a_\ell^m$*. If additionally* $a_2^1 \neq 0$*, the residual* $\mathrm{SO}(2)_z$ *freedom is consumed by choosing* φ *such that* $\mathrm{Im}(a_2^1) = 0$*, i.e.* $a_2^1 \in \mathbb{R}_{>0}$*.*

Proof. The $\ell = 1$ Wigner D-matrix $D^1(g)$ is the standard 3×3 rotation matrix. Since $\mathrm{SO}(3)$ acts transitively on S^2, any nonzero $\mathbf{F}_1$ can be rotated to align with $\hat{z}$, leaving an $\mathrm{SO}(2)_z$ stabiliser. Choosing $\varphi = -\arg(a_2^1)$ yields $a_2^1 \in \mathbb{R}_{>0}$, exhausting all three rotational degrees of freedom. □

Lemma 17 (Recovery of $\mathbf{F}_1$)**.** *Given* a_0^0 *from Lemma 15, the entry* $\beta_{0,1,1}$ *determines* $\|\mathbf{F}_1\|$*. Combined with the gauge choice of Lemma 16, this uniquely determines* $\mathbf{F}_1$*.*

Proof. By (3) with $(\ell_1, \ell_2, \ell) = (0, 1, 1)$:

$$\beta_{0,1,1} = a_0^0 \sum_m |a_1^m|^2 = a_0^0\, \|\mathbf{F}_1\|^2,$$

using $C^{1,\,m}_{0,\,0;\,1,\,m} = 1$ (trivial coupling). Since a_0^0 is known and nonzero, $\|\mathbf{F}_1\|^2 = \beta_{0,1,1}/a_0^0$. After gauge fixing, $\mathbf{F}_1 = (0, \|\mathbf{F}_1\|, 0)^\top$. □

Lemma 18 (Joint seed recovery of $\mathbf{F}_2$ and $\mathbf{F}_3$)**.** *After gauge fixing,* $(\mathbf{F}_2, \mathbf{F}_3)$ *are jointly and generically determined by a constant-size seed set* $\mathcal{S}_{\leq 3}$ *of bispectral triples involving only degrees* ≤ 3*.*

Proof. After gauge fixing, $\mathbf{F}_1 = (0, c, 0)^\top$ with $c = \|\mathbf{F}_1\|$, and $\mathrm{Im}(a_2^1) = 0$. The real-signal constraint gives $\mathbf{F}_2$ four real unknowns ($a_2^0,\ a_2^1 \in \mathbb{R},\ \mathrm{Re}(a_2^2),\ \mathrm{Im}(a_2^2)$) and $\mathbf{F}_3$ seven real unknowns ($a_3^0,\ a_3^1 \in \mathbb{R},\ \mathrm{Re}(a_3^m),\ \mathrm{Im}(a_3^m)$ for $m = 2, 3$), for a total of 11 real unknowns.

Why $\ell = 2$ *cannot be solved in isolation.* The chain entry $\beta_{1,1,2}$ determines a_2^0 via

$$\beta_{1,1,2} = C^{2,\,0}_{1,\,0;\,1,\,0}\, c^2\, \overline{a_2^0} = \sqrt{\tfrac{2}{3}}\, c^2\, a_2^0. \tag{4}$$

After fixing a_2^0, three real unknowns remain. However, the available $\ell = 2$ self-coupling channels are insufficient: $\beta_{2,2,0} \propto \|\mathbf{F}_2\|^2$ is redundant with the power entry $\beta_{0,2,2}$, and $\beta_{2,2,1}$ vanishes identically because the $2 \otimes 2 \to 1$ Clebsch–Gordan channel is antisymmetric under exchange of the two degree-2 factors, while the product $a_2^{m_1} a_2^{m_2}$ is symmetric:

$$(\mathbf{F}_2 \otimes \mathbf{F}_2)_1 = 0.$$

This leaves only two genuinely informative $\ell = 2$ scalars ($\beta_{1,1,2}$ and $\beta_{0,2,2}$, with $\beta_{2,2,2}$ as a third cubic entry), which cannot determine four real unknowns.

Joint seed set. We form the seed set $\mathcal{S}_{\leq 3}$ from the informative $\ell = 2$ entries and all $\ell = 3$ entries:

$$\begin{aligned} &\text{from } \ell = 2: \quad \beta_{1,1,2} \text{ (chain)},\ \beta_{1,2,1} \text{ (cross)},\ \beta_{0,2,2} \text{ (power)}, \\ &\text{from } \ell = 3: \quad \beta_{2,2,3},\ \beta_{1,2,3} \text{ (chain)},\ \beta_{2,3,2},\ \beta_{2,3,1},\ \beta_{1,3,2} \text{ (cross)},\ \beta_{0,3,3} \text{ (power)},\ \beta_{3,3,2} \text{ (self)}. \end{aligned}$$

This gives 10 scalar equations. The $\ell = 3$ chain and cross entries that involve $\mathbf{F}_2$ (e.g. $\beta_{1,2,3}$, $\beta_{2,3,1}$) couple the two degrees, providing the missing constraints on $\mathbf{F}_2$ that the isolated $\ell = 2$ system lacked.

Generic invertibility. Define the seed map $\Phi\colon (\mathbf{F}_2, \mathbf{F}_3) \mapsto \{\beta_\tau\}_{\tau \in \mathcal{S}_{\leq 3}}$. Its Jacobian J_Φ is an 10×11 real matrix whose entries are polynomials in the 11 unknowns (with CG coefficients as fixed weights). We exhibit a witness point—a randomly sampled generic signal—at which a 10×10 minor of J_Φ has nonzero determinant (verified symbolically). Since this determinant is a nonzero polynomial over $\mathbb{R}$, its zero set is a proper algebraic subvariety of Lebesgue measure zero. Therefore Φ is a submersion at generic points, and $(\mathbf{F}_2, \mathbf{F}_3)$ are locally determined by the seed bispectrum up to a discrete (generically finite) ambiguity. □

Lemma 19 (Linear bootstrap for $\ell \geq 4$)**.** *For each $\ell \geq 4$, the chain and cross entries provide at least $2\ell + 1$ linear equations in $\mathbf{F}_\ell$, and the coefficient matrix has rank $2\ell + 1$ for Lebesgue-almost-every signal.*

Proof. We work on the complexified irreducible modules $V_j := \mathrm{Sym}^{2j}(\mathbb{C}^2)$, with monomial basis $e_k^{(j)} := x^k y^{2j-k}$, $k = 0, \ldots, 2j$. The real signal space is Zariski-dense in this complexification, so it is enough to prove that the relevant maximal minor is not the zero polynomial on the complexified coefficient space.

Finite low degrees. For $4 \leq \ell \leq 7$, the selector uses explicit linear blocks $\mathcal{T}_\ell^{\mathrm{small}}$ (for $\ell = 4, 5, 6, 7$) and the same closed-form family below takes over at $\ell = 8$. For each of these four degrees, we evaluate the resulting $(2\ell + 1) \times (2\ell + 1)$ matrix at an explicit rational witness and compute its determinant exactly by symbolic algebra; in every case the determinant is nonzero. Hence the lemma holds for $4 \leq \ell \leq 7$.

Uniform family for $\ell \geq 8$. For $\ell \geq 8$, use the $2\ell + 1$ rows

$$\begin{aligned} X_a^{(0)} &:= (a, \ell, \ell - a), \qquad 1 \leq a \leq \ell - 1, \\ X_a^{(1)} &:= (a, \ell, \ell - a + 1), \qquad 2 \leq a \leq \ell - 1, \\ C_a &:= (a, \ell - a, \ell), \qquad 1 \leq a \leq 4. \end{aligned}$$

By Proposition 10, this is exactly the linear budget $2\ell + 1$.

Binary-form witness. Fix a formal parameter t and choose lower-degree forms

$$\begin{gathered} f_1 = x^2, \qquad f_2 = x^4 + t\, x^3 y, \qquad f_3 = x^6 + t^2 x y^5, \qquad f_4 = x^8 + t^3 y^8, \\ f_{\ell-4} = x^{2\ell-8} + t^4 y^{2\ell-8}, \qquad f_{\ell-3} = x^{2\ell-6} + t^5 y^{2\ell-6}, \end{gathered}$$

and $f_j = x^{2j}$ for all remaining $j < \ell$. Chain rows are obtained by multiplication $V_a \otimes V_{\ell-a} \to V_\ell$. Cross rows $X_a^{(0)}$ and $X_a^{(1)}$ are, respectively, the $2a$-th and $(2a - 1)$-st transvectants in the binary-form model.

Interior block. For $5 \leq a \leq \ell - 1$, both factors in $X_a^{(0)}$ are pure monomials, so the $2a$-th transvectant gives an exact monomial selector:

$$X_a^{(0)} = (2a)!^2\, (e_{2\ell-2a}^{(\ell)})^*.$$

Likewise, for $6 \leq a \leq \ell - 1$,

$$X_a^{(1)} = (2a)!(2a - 1)!\, (e_{2\ell-2a+1}^{(\ell)})^*.$$

Hence these rows already form a diagonal block on the interior columns

$$2, 3, \ldots, 2\ell - 11, 2\ell - 10.$$

All corresponding diagonal entries are nonzero factorials.

Boundary block. The remaining 12 columns are

$$0,\ 1,\ 2\ell-9,\ 2\ell-8,\ 2\ell-7,\ 2\ell-6,\ 2\ell-5,\ 2\ell-4,\ 2\ell-3,\ 2\ell-2,\ 2\ell-1,\ 2\ell.$$

They are handled by the rows

$$C_1,\ C_2,\ C_3,\ C_4,\ X_1^{(0)},\ X_2^{(0)},\ X_3^{(0)},\ X_4^{(0)},\ X_2^{(1)},\ X_3^{(1)},\ X_4^{(1)},\ X_5^{(1)}.$$

Directly from the multiplication and transvectant formulas, the leading t-terms of this 12×12 boundary matrix are diagonal except for the two rows $(C_4, X_4^{(0)})$ against the two columns $(e_0^{(\ell)}, e_{2\ell-8}^{(\ell)})$. The relevant leading 2×2 block is

$$\begin{pmatrix} 1 & 1 \\ 8!\,(2\ell)(2\ell-1)\cdots(2\ell-7) & (8!)^2 \end{pmatrix},$$

whose determinant equals

$$8!\,\Big((8!) - (2\ell)(2\ell-1)\cdots(2\ell-7)\Big) \neq 0$$

for every $\ell \geq 10$. All other boundary diagonal entries are explicit nonzero factorial coefficients: 1 for C_1, C_2, C_3 and $(2a)!^2$ or $(2a)!(2a-1)!$ for the cross rows. Therefore the boundary block has a nonzero leading determinant term.

Combining the interior diagonal block and the non-singular boundary block, we obtain a $(2\ell+1)\times(2\ell+1)$ minor $A'_\ell(t)$ with

$$\det A'_\ell(t) = c_\ell\, t^{q_\ell} + O(t^{q_\ell+1}), \qquad c_\ell \neq 0.$$

Hence $\det A'_\ell$ is not the zero polynomial. Therefore the rank-deficient set is a proper algebraic subvariety of codimension at least one, so A'_ℓ has full rank $2\ell+1$ for Lebesgue-almost-every signal. At a generic point, $\mathbf{F}_\ell = A^\dagger \mathbf{b}$. □

Proof of Theorem 14. We proceed by induction on $\ell = 0, 1, \ldots, L$.

Base cases. Lemma 15 recovers $\mathbf{F}_0 = (a_0^0)$. Lemmas 17 and 16 recover $\mathbf{F}_1$ and fix the rotation gauge, consuming all three $\mathrm{SO}(3)$ degrees of freedom. Lemma 18 jointly recovers $(\mathbf{F}_2, \mathbf{F}_3)$ via the seed set $\mathcal{S}_{\leq 3}$.

Inductive step ($\ell \geq 4$). Suppose $\mathbf{F}_0, \ldots, \mathbf{F}_{\ell-1}$ are known (up to the common gauge). By Lemma 19, the chain and cross entries at degree ℓ form a linear system $A\,\mathbf{F}_\ell = \mathbf{b}$ of rank $2\ell+1$ at generic points, uniquely determining $\mathbf{F}_\ell$.

Conclusion. The sequence $\mathbf{F}_0, \ldots, \mathbf{F}_L$ is recovered up to the rotation $g \in \mathrm{SO}(3)$ fixed in Lemma 16. By Plancherel's theorem, f is recovered from its coefficients. Therefore, $\beta_{\text{sel}}(f) = \beta_{\text{sel}}(f')$ implies $f' \in \mathrm{SO}(3)\cdot f$ for generic f. □

A.5 Comparison with existing results

Table 9: Comparison of bispectral invariants for signals on S^2 or finite groups.

Result	Invariant	Size	Complete?
Kakarala [Kakarala, 1992]	Full bispec.	$O(L^3)$	Yes (constr.)
Edidin & Satriano [Edidin and Satriano, 2023]	Full bispec. (band-lim.)	$O(L^3)$	Yes (generic)
Bendory et al. [Bendory et al., 2022]	Freq. marching	$O(L^2)$/shell	Yes (3 shells)
Mataigne et al. [Mataigne et al., 2024]	Sel. G-bispec. (finite)	$O(\lvert G\rvert)$	Yes (exact)
This work	**Sel.** $\mathrm{SO}(3)$ **on** S^2	$\boldsymbol{O(L^2)}$	**Yes (generic)**

A.6 Complexity in pixel space

On S^2, a band-limited signal at degree L is sampled on an equiangular grid of size $N_\theta \times N_\varphi \approx 2L \times 2L$, so $P = 4L^2$ and $L = \Theta(\sqrt{P})$.

Output size.

Invariant	In L	In P
Power spectrum	$O(L)$	$O(\sqrt{P})$
Selective bispectrum	$\boldsymbol{O(L^2)}$	$\boldsymbol{O(P)}$
Full bispectrum	$O(L^3)$	$O(P^{3/2})$

The selective bispectrum is *linear* in the number of pixels.

Computation cost.

Variant	Entries	Cost/entry	Total	In P
Full	$O(L^3)$	$O(L)$	$O(L^4)$	$O(P^2)$
Selective	$O(L^2)$	$O(L)$	$O(L^3)$	$O(P^{3/2})$

Concrete example. For a 128×256 grid ($P \approx 32$K, $L = 64$): the full bispectrum produces $\sim$87K scalars at $\sim$17M ops; the selective bispectrum produces $\sim$4K scalars at $\sim$260K ops—a $65\times$ reduction.

A.7 Relation to the inversion gap

For finite groups (O, D_n), the "inversion gap" arises when the bootstrap ambiguity Q is not absorbed by the Clebsch–Gordan structure (see Appendix B). For $\mathrm{SO}(3)$ on S^2, the situation differs fundamentally:

1. The ambiguity in $\mathbf{F}_1$ is an $\mathrm{SO}(3)$ rotation g (3 DOF).
2. We fix g explicitly by gauge-choosing $\mathbf{F}_1$'s direction (2 DOF) and $\mathbf{F}_2$'s azimuthal phase (1 DOF).
3. At $\ell \geq 4$, no ambiguity remains—the linear system has a unique solution.

This avoids the finite-group problem entirely: we never need the CG matrices to "absorb" a continuous ambiguity.

A.8 Limitations and open questions

1. **Genericity assumption.** The proof requires f to avoid a measure-zero algebraic variety (Assumption 12). Characterising this variety explicitly would strengthen the result.
2. **Numerical conditioning.** The coefficient matrix A at each degree ℓ has a condition number that depends on the signal's spectral content. A stability analysis bounding $\kappa(A)$ in terms of the spectral decay rate is an open problem.
3. **Inversion algorithm.** The forward selective bispectrum is implemented. Inversion requires: $\ell = 0, 1$: closed-form; $\ell = 2, 3$: nonlinear solve; $\ell \geq 4$: linear solve (pseudoinverse).

B Octahedral Inversion Gap – Full Analysis

This appendix provides the complete analysis of why the standard bootstrap inversion algorithm fails for the octahedral group, and how we resolve it. The material is complementary to Section 3.5 in the main text.

B.1 The bootstrap strategy (recap)

All inversion algorithms in Mataigne et al. [2024] share the same skeleton:

1. Recover $\hat{f}(\rho_0)$ from $\beta_{\rho_0,\rho_0} = \hat{f}(\rho_0)^3$.
2. Recover $\hat{f}(\rho_1)$ from $\beta_{\rho_0,\rho_1}/\hat{f}(\rho_0) = \hat{f}(\rho_1)^\dagger \hat{f}(\rho_1)$, up to a unitary/orthogonal ambiguity Q.
3. Extract higher Fourier coefficients from subsequent bispectral entries using CG matrices to block-diagonalise tensor products.

Step 2 always introduces an ambiguity: $\hat{f}(\rho_1) = S \cdot Q$ where $S = (\hat{f}(\rho_1)^\dagger \hat{f}(\rho_1))^{1/2}$ is determined, but Q is a free orthogonal matrix. For the algorithm to work, this ambiguity must not corrupt step 3.

B.2 Why it works for C_n

The irreps are 1D: $\rho_k(g) = e^{2\pi i k g/n}$. The ambiguity in $\hat{f}(\rho_1)$ is a phase $e^{i\varphi}$, $\varphi \in [0, 2\pi)$. Step 3 computes:

$$\hat{f}(\rho_{k+1}) = \left(\frac{\beta_{\rho_1,\rho_k}}{\hat{f}(\rho_1)\hat{f}(\rho_k)} \right)^\dagger .$$

The phase $e^{i\varphi}$ propagates coherently through all coefficients as $e^{i\varphi k}$, corresponding to a cyclic shift $\Theta(g) \to \Theta(g + \varphi)$. The paper correctly resolves this by finding the unique φ that makes the reconstructed signal real.

B.3 Why it works for D_n

The generating irrep ρ_1 is 2D. Recovery gives $\hat{f}(\rho_1) = V\Lambda^{1/2}V^\dagger U$ with $U \in \mathrm{O}(2)$. Step 3 extracts higher coefficients via:

$$\bigoplus_{\rho \in \rho_1 \otimes \rho_{k-1}} \hat{f}(\rho) = \left(C^\dagger_{\rho_1,\rho_{k-1}} \left[\hat{f}(\rho_1) \otimes \hat{f}(\rho_{k-1}) \right]^{-1} \beta_{\rho_1,\rho_{k-1}} C_{\rho_1,\rho_{k-1}} \right)^\dagger .$$

The 2D irreps of D_n are restrictions of $\mathrm{SO}(2)$ representations to $\mathbb{Z}/n\mathbb{Z}$. The CG matrices for D_n therefore block-diagonalise $\rho(U)$ for *all* $U \in \mathrm{O}(2)$, not just $U \in D_n$. Concretely, $C^\dagger(U \otimes U)C$ remains block-diagonal for any $U \in \mathrm{O}(2)$. The $\mathrm{O}(2)$ ambiguity passes cleanly through the CG extraction.

B.4 Why it fails for O

For the octahedral group, ρ_1 is the standard 3D representation. From β_{ρ_0,ρ_1} we recover $\hat{f}(\rho_1)^\top \hat{f}(\rho_1) = \beta_{\rho_0,\rho_1}/\hat{f}(\rho_0)$. Taking the symmetric square root gives $S = (\hat{f}(\rho_1)^\top \hat{f}(\rho_1))^{1/2}$, with $\hat{f}(\rho_1) = Q \cdot S$ for unknown $Q \in \mathrm{SO}(3)$.

The next step extracts $\hat{f}(\rho_2)$ and $\hat{f}(\rho_3)$ from β_{ρ_1,ρ_1} using the CG matrix C_{11} that block-diagonalises $\rho_1 \otimes \rho_1$:

$$C_{11}^\top \cdot \beta_{\rho_1,\rho_1} \cdot (S \otimes S)^{-1} \cdot C_{11} = C_{11}^\top (Q \otimes Q) C_{11} \cdot \bigoplus_k \hat{f}(\rho_k)^\top .$$

The CG matrix C_{11} block-diagonalises $\rho_1(g) \otimes \rho_1(g)$ for $g \in O$:

$$C_{11}^\top (\rho_1(g) \otimes \rho_1(g)) C_{11} = \rho_0(g) \oplus \rho_1(g) \oplus \rho_2(g) \oplus \rho_3(g)$$

with dimensions $1+3+3+2 = 9$. But Q is **not** constrained to be an element of O. It is an arbitrary element of $\mathrm{SO}(3)$, which is a continuous group with 3 parameters, while O has only 24 elements. For a generic $Q \in \mathrm{SO}(3) \setminus O$, the matrix $C_{11}^\top (Q \otimes Q) C_{11}$ is **not block-diagonal**.

Cross-block leakage. The Kronecker decomposition of $\rho_1 \otimes \rho_1$ for O gives blocks:

Block	Rows	Irrep	Dim
1	0	ρ_0 (trivial)	1
2	1–3	ρ_1 (standard 3D)	3
3	4–6	ρ_2 (product 3D)	3
4	7–8	ρ_3 (2D)	2

When $Q = \rho_1(g)$ for $g \in O$, the off-diagonal blocks (e.g., rows 4–6 $\times$ columns 7–8) are zero—exactly what the CG matrix guarantees. When Q is a generic rotation, e.g., $Q = R_x(0.7)$, then $|[C_{11}^\top (Q \otimes Q) C_{11}]_{4:7,\,7:9}|_\infty \approx 0.788$. This is not a small perturbation—it is $O(1)$. The recovered $\hat{f}(\rho_2)$ is a mixture of ρ_2 and ρ_3 contributions.

Why this doesn't happen for D_n. The 2D irreps of D_n are restrictions of $\mathrm{SO}(2)$ irreps. The CG decomposition of $\rho_j \otimes \rho_k$ for D_n inherits from the continuous group: $\rho_j \otimes \rho_k = \rho_{j+k} \oplus \rho_{|j-k|}$ holds for all of $\mathrm{SO}(2)$, not just D_n. The CG matrices are inherited from $\mathrm{SO}(2)$ and respect arbitrary rotations. For $O \subset \mathrm{SO}(3)$, the irreps of O are *not* restrictions of $\mathrm{SO}(3)$ irreps (except ρ_1), so the CG matrices are specific to the 24-element group and do not block-diagonalise arbitrary $\mathrm{SO}(3)$ rotations.

B.5 Connection to the paper's own remarks

Mataigne et al. [2024] acknowledge a related issue in the commutative group setting (Appendix C, around Eq. 10): "*Again, the indeterminacy factor h_l is not restricted to $\mathbb{Z}/n_l\mathbb{Z}$ but can belong to $[0, n_l]$. We will have to solve this issue further.*" This is the same phenomenon in a different guise. For commutative groups, the continuous indeterminacy is scalar (a phase) and can be resolved by enforcing real-valuedness. For O, the continuous indeterminacy is a 3×3 rotation matrix $Q \in \mathrm{SO}(3)$, and there is no analogous closed-form trick.

B.6 Working alternative

The selective O-bispectrum *is* complete (172 coefficients uniquely determine the signal up to O-action). The issue is purely algorithmic. We implement a working inversion using:

1. **Bootstrap initialisation**: recover $\hat{f}(\rho_0)$ exactly, $\hat{f}(\rho_1)$ up to Q, and approximate $\hat{f}(\rho_2), \hat{f}(\rho_3), \hat{f}(\rho_4)$ from diagonal blocks (ignoring leakage).
2. **Levenberg–Marquardt correction**: use forward-mode autodiff through the bispectrum computation to solve $\|\beta(f_{\mathrm{rec}}) - \beta_{\mathrm{target}}\|^2 = 0$ via damped least-squares steps. Each step is a single linear solve, not iterative optimisation. Jacobians are computed via `torch.func.jacfwd` composed with `torch.vmap` for batched forward-mode AD.
3. **Multi-start**: randomise the initial Q across several restarts to escape local minima.

This converges reliably in <50 iterations and is fully differentiable.

REVIEW: #11 — The main text (Sec. 3.5) says "convergence plots, is in Appendix B" but this appendix has no convergence figure or success-rate table. Add a compact quantitative table: inversion success rate, median iterations, worst-case iterations, runtime, and a few failure cases. This is one of the claimed technical contributions—it needs quantitative evidence.

B.7 Summary

Table 10: Bootstrap inversion across group families.

Group	Irrep dim	Ambiguity	CG absorbs?	Bootstrap?
C_n	1	$e^{i\varphi} \in \mathrm{SO}(2)$	N/A (scalar)	Yes
D_n	2	$U \in \mathrm{O}(2)$	Yes (CG from $\mathrm{SO}(2)$)	Yes
O	3	$Q \in \mathrm{SO}(3)$	**No**	**No**

B.8 Numerical disproof of CG absorption for O

The following code demonstrates the cross-block leakage:

```python
import torch
from bispectrum.o_on_r3 import OonR3

bsp = OonR3()
C11 = bsp._cg_11.to(torch.float64)

# Generic SO(3) rotation (not in O)
theta = 0.7
Q = torch.tensor([
    [1.0, 0.0, 0.0],
    [0.0, torch.cos(torch.tensor(theta)),
     -torch.sin(torch.tensor(theta))],
    [0.0, torch.sin(torch.tensor(theta)),
      torch.cos(torch.tensor(theta))],
], dtype=torch.float64)

QQ = torch.kron(Q, Q)
M = C11.T @ QQ @ C11

# rho2 block: rows 4:7, rho3 block: rows 7:9
cross = M[4:7, 7:9]
print(f"Cross-block max: {cross.abs().max():.6f}")
# Output: ~0.788 -- far from zero

# Compare with a group element
rho1_g4 = bsp._get_irrep_mats(1)[4].to(torch.float64)
QQ_grp = torch.kron(rho1_g4, rho1_g4)
M_grp = C11.T @ QQ_grp @ C11
cross_grp = M_grp[4:7, 7:9]
print(f"Cross-block for g in O: {cross_grp.abs().max():.6f}")
# Output: ~0.0 (machine precision)
```

C Extended Benchmark Results

This appendix supplements the summary in Table 5 (main text) with scaling plots produced by `benchmarks/benchmark.py` on a single NVIDIA H100 80 GB GPU (PyTorch 2.10).

C.1 Forward-pass timing

The selective forward pass is dominated by kernel-launch overhead rather than arithmetic, making it effectively free relative to backbone computation. The full bispectrum exhibits the expected super-linear growth and becomes impractical beyond $|G| \approx 256$ for C_n.

C.2 GPU throughput and inversion

All modules exhibit linear throughput scaling with batch size, confirming that the implementation is memory-bandwidth-bound rather than compute-bound at these scales.

D PCam: Extended Results

D.1 Published baselines

Table 11 places our results in the context of published methods on PCam and the closely related Camelyon16 benchmark. Our models operate at 68K–920K parameters and are trained on only 10% of the data.

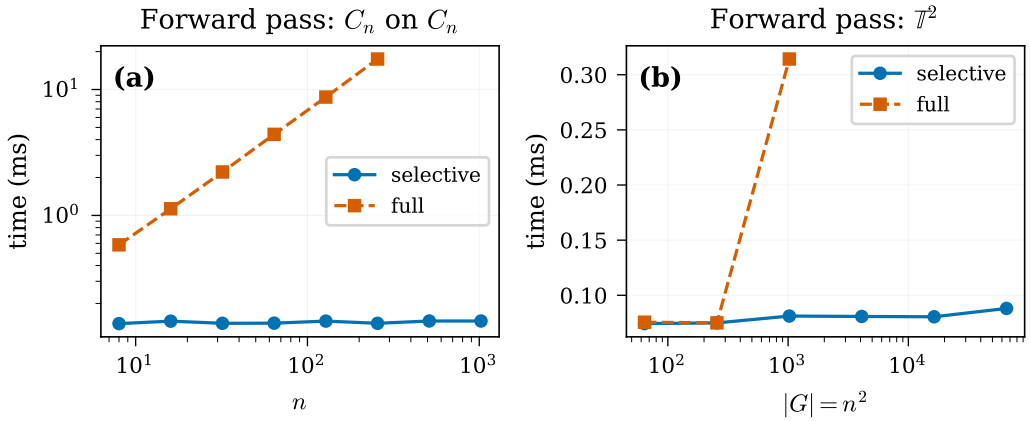

Figure 5: Forward-pass wall-clock time (batch = 16, H100 GPU). **(a)** C_n: the selective pass stays flat at ∼0.14 ms regardless of n, while the full bispectrum grows super-linearly (17 ms at n=256). **(b)** $\mathbb{T}^2$: selective is constant at ∼0.08 ms up to $|G|$=65,536.

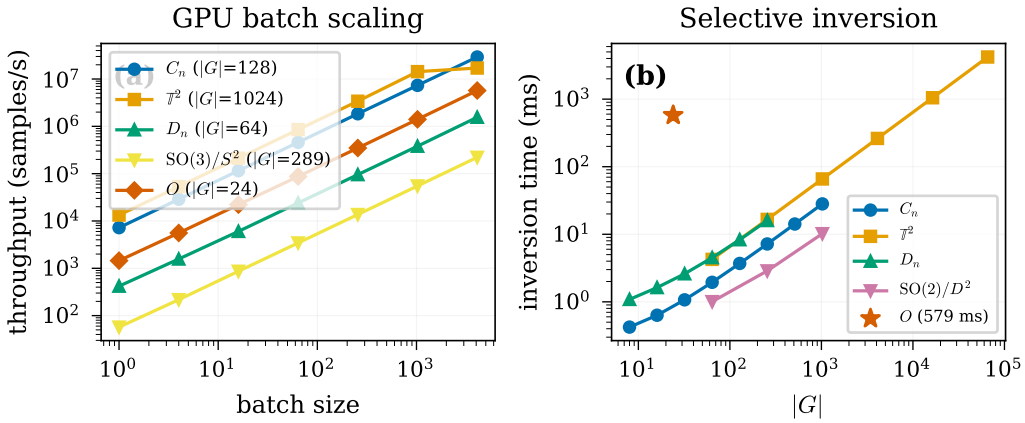

Figure 6: **(a)** GPU throughput (samples/s) vs. batch size for the selective forward pass. C_n and $\mathbb{T}^2$ exceed 10^7 samples/s; the octahedral module reaches 5.7×10^6 samples/s. All modules scale linearly with batch size. **(b)** Inversion wall-clock time. Abelian groups (C_n, D_n, SO(2)/disk) scale roughly linearly. The octahedral group requires ∼580 ms due to Levenberg–Marquardt optimisation over 5 irreducible representations. SO(3) on S^2 inversion is not yet implemented.

D.2 Model variants

All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone with block configuration $(4, 4, 4)$, 24 initial channels, and compression factor 0.5, following the architecture of Veeling et al. [2018] but replacing the final invariant map. Six model variants are evaluated:

- **Standard**: vanilla CNN with random rotation/reflection augmentation and global average pooling. No group dimension, so ∼8× fewer parameters at matched growth rate.
- **NormReLU**: C_8-equivariant backbone with norm-based nonlinearity and max pooling over the group dimension.
- **Gated**: C_8-equivariant backbone with gated nonlinearity (doubles internal channels) and group max pool.

Table 11: Published baselines on PatchCamelyon from Veeling et al. [2018], trained on 100% data.

Method	Reference	Params	ACC	AUC
P4M-DenseNet (C_8)	Veeling et al. 2018	119K	0.898	0.963
P4M-DenseNet M	Veeling et al. 2018	19K	0.893	0.958
DenseNet (augmented)	Veeling et al. 2018	128K	0.881	0.951
DenseNet (no aug.)	Veeling et al. 2018	128K	0.876	0.955

- **Fourier-ELU**: C_8-equivariant backbone with Fourier-space ELU nonlinearity and group max pool.
- **Bispectrum**: C_8-equivariant backbone with ReLU nonlinearity and selective G-bispectrum via `CnonCn`(n=8) as the terminal invariant map.
- **Disk bispectrum**: no convolutional backbone; `SO2onDisk` is applied per RGB channel to the raw 96×96 patches, producing rotation-invariant Fourier–Bessel bispectrum coefficients that are fed into a small MLP classifier.

D.3 Disk bispectrum as a backbone-free baseline

The `SO2onDisk` model applies the selective disk bispectrum directly to the raw input patches without any learned convolutional features. At 100K parameters (bandlimit $= 10$) it reaches AUC 0.794—substantially below the learned models, as expected for a handcrafted feature extractor. However, its rotation robustness is near-perfect: $\sigma_{\text{AUC}} < 2 \times 10^{-5}$ across 12 test-time rotation angles, compared to residual angle-dependent variation in the C_8 models. At 1% training data, the disk bispectrum retains AUC 0.778, showing that the analytically rotation-invariant features degrade gracefully when data is scarce. This variant demonstrates that `SO2onDisk` can serve as a drop-in rotation-invariant feature extractor from the library, independent of any equivariant backbone.

D.4 Training details

To enable a controlled Pareto comparison, we sweep the DenseNet growth rate $k \in \{3, 4, 6, 8, 12\}$ for equivariant models (and $k \in \{6, 12, 20, 30, 35\}$ for the standard CNN to cover a comparable parameter range), yielding models from $\sim$30K to $\sim$1.6M parameters. All models are trained on **10% of training data** (26,214 images) with AdamW [Loshchilov and Hutter, 2019] (lr $= 10^{-3}$, weight decay 10^{-4}), cosine warm restarts, mixed-precision training, and early stopping on validation AUC (patience 10, max 50 epochs).

D.5 Pareto analysis interpretation

Architectural stability. The bispectrum's AUC varies by only $\sim$1pp across a $12\times$ range of model sizes (80K–920K parameters), while norm swings by $>$7pp (0.878–0.954). For practitioners, bispectral pooling "just works" without growth rate tuning.

Matched parameters. At matched parameters ($\sim$100K), most equivariant models outperform the standard CNN. Norm peaks at 0.954 AUC (k=4, 110K params), followed by Fourier-ELU (0.937), bispectrum (0.933), standard (0.928), and gate (0.924). The gate variant slightly underperforms the standard baseline at this operating point.

Scaling behavior. As parameters increase from the smallest to the largest configuration, standard, norm, and Fourier-ELU all degrade (by 1.8, 2.6, and 0.9 AUC points respectively), while the bispectrum improves (+0.9 points, reaching 0.943 at 920K params). This suggests that the bispectrum's complete invariant map scales more gracefully with capacity.

Data scaling. All models improve substantially from 1% to 10% data. The bispectrum shows the largest absolute gain (+6.8pp), consistent with the hypothesis that complete invariants benefit more from additional data.

E OrganMNIST3D: Extended Results

E.1 Published baselines

Table 12 situates our results against published methods on the same benchmark. Our models are deliberately small ($\leq$463K parameters) to enable controlled ablation of the pooling mechanism. The relevant comparison is against equivariant methods at a similar scale.

Table 12: Published baselines on OrganMNIST3D for context. Accuracy differences with respect to large-scale models reflect the deliberate parameter constraint of our ablation study.

Method	Reference	Params	ACC	AUC
ResNet-18 + 3D	Yang et al. 2023	33M	0.907	0.996
ResNet-18 + ACS	Yang et al. 2023	11M	0.900	0.994
ILPOResNet-50	Zhemchuzhnikov and Grudinin [2024]	38K	0.879	0.992
EquiLoPO (local train.)	Zhemchuzhnikov and Grudinin [2025]	418K	0.866	0.991
SE3MovFrNet	Sangalli et al. [2023]	—	0.745	—
Regular SE(3) conv	Kuipers and Bekkers [2023]	172K	0.698	—

E.2 Rotation analysis

REVIEW: #6 — CRITICAL: Reviewer says Figure 4 shows 38.6, 56.8, 52.6, 57.5 for original/rotated test sets, but Table 6 in the main text shows 60.1, 56.8, 73.0, 72.6. Audit `organ3d_analysis.pdf` against the raw logs. If the figure was generated from a different run or seed, regenerate from the same data as Table 6.

Figure 7 shows that all equivariant models achieve identical performance on the original and octahedrally rotated test sets ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades under rotation.

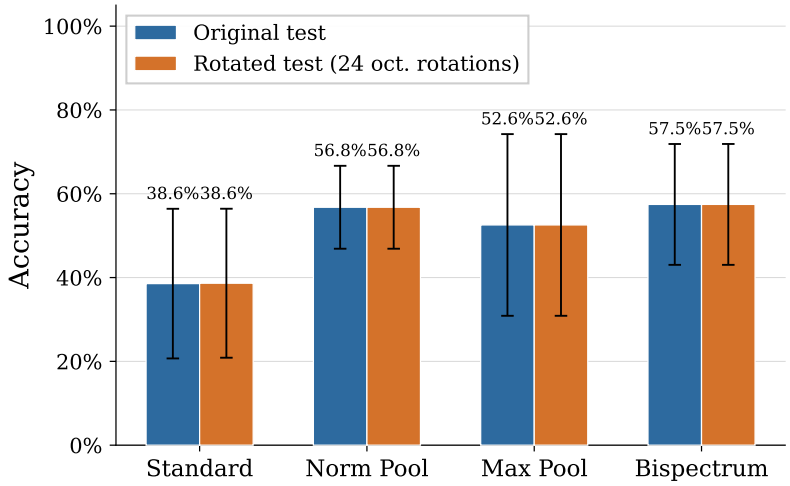

Figure 7: OrganMNIST3D test accuracy on the original and octahedrally rotated test sets. Error bars show standard deviation over three seeds. All equivariant models achieve identical performance under rotation ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades.

E.3 Wider channels

REVIEW: #7 — Table 12 caption says "mean $\pm$ std over 3 seeds" but the $(8, 16)$ and $(16, 32)$ rows have no std devs. Either re-run those configs with 3 seeds and add stds, or fix the caption to say they are single-run.

To test whether the advantage of bispectral completeness persists at higher capacity, we run max pool and bispectrum at channel widths $(8, 16)$ and $(16, 32)$ with full training data.

Table 13: OrganMNIST3D accuracy at wider channel configurations (mean $\pm$ std over 3 seeds, full training data).

Channels	Params	Max Pool	Bispectrum
$(4, 8)$	$\sim$375K / 463K	$.730 \pm .033$	$.726 \pm .027$
$(8, 16)$	$\sim$1.5M	.749	.753
$(16, 32)$	$\sim$6.3M	**.830**	.715

At $(8, 16)$ the two methods are equivalent (75.3% bispectrum vs. 74.9% max pool, both at $\sim$1.5M parameters). At $(16, 32)$ max pool reaches 83.0% while bispectrum drops to 71.5%—the 6.3M parameter bispectrum model likely overfits the 971 training samples. This suggests that the bispectrum's completeness advantage is most pronounced in the data-limited rather than the parameter-limited regime, consistent with the data efficiency findings in the main text.

F Experimental Details

F.1 Hyperparameters

Table 14 summarises the training hyperparameters for all three experimental settings.

Table 14: Training hyperparameters across experiments.

	PCam	OrganMNIST3D	Sph. MNIST
Optimizer	AdamW	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}	10^{-4}
Batch size	64–256	16–64	256
Scheduler	Cosine warm restarts	—	—
Max epochs	50	100	50
Early stopping	Val AUC, pat. 10	Val AUC, pat. 15	Val ACC, pat. 10
Seeds	42†	42, 123, 456	42, 123, 456
Mixed precision	Yes	No	No

†Pareto sweep uses seed 42; data efficiency uses seeds 42, 123, 456.

F.2 Hardware

All experiments were run on a single NVIDIA H100 GPU (80 GB). Training wall-clock times range from $\sim$5 minutes (OrganMNIST3D, 971 samples) to $\sim$20 minutes (PCam, 26K samples at 10% data). Spherical MNIST bispectrum computation (forward pass through `SO3onS2` at $\ell_{\max} = 15$) takes $\sim$0.3s per batch of 64 on GPU.

F.3 Data preprocessing

PCam. PatchCamelyon [Veeling et al., 2018] consists of 327,680 RGB patches of size 96×96 pixels extracted from whole-slide images of lymph node sections [Ehteshami Bejnordi et al., 2017]. We use the standard train/val/test split. Images are loaded from HDF5 files and normalised to $[0, 1]$. The standard (non-equivariant) baseline uses random rotation and reflection augmentation during training; all equivariant models are trained without augmentation.

OrganMNIST3D. OrganMNIST3D [Yang et al., 2023] contains 1,742 abdominal CT organ volumes of size $28 \times 28 \times 28$ voxels with 11 organ classes (train/val/test = 971/161/610). Volumes are normalised to $[0, 1]$. The standard baseline uses random octahedral augmentation (one of 24 rotations per sample); equivariant models are trained without augmentation. Kernel rotations are implemented as exact voxel permutations on the 3^3 grid, avoiding interpolation artifacts.

Spherical MNIST. MNIST digits are projected onto the sphere via stereographic projection following Cohen et al. [2018] and sampled on a 64×128 equiangular grid. The selective bispectrum is

computed at $\ell_{\max} = 15$, producing 256 complex invariant coefficients (512 real features after taking log-magnitude), which are fed to a 3-layer MLP: Linear(512, 256)–BN–ReLU–Linear(256, 128)–BN–ReLU–Linear(128, 10). The power spectrum MLP uses the same structure with hidden=128: Linear(16, 128)–BN–ReLU–Linear(128, 64)–BN–ReLU–Linear(64, 10). The standard CNN has 4 convolutional layers (16–32–64–64 channels) followed by adaptive average pooling and a Linear(1024, 128)–ReLU–Linear(128, 10) classifier.

F.4 Aggregation procedures

PCam. The Pareto analysis (Figure 2, Table 6) uses a single seed (42) to reduce computational cost across the full growth-rate sweep. The data efficiency comparison (1% vs. 10%) uses 3 seeds per configuration.

OrganMNIST3D. All results in Table 7 report mean $\pm$ std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is the standard deviation of test accuracy across all 24 octahedral rotations of the test set. The data efficiency analysis (Table 5 in main text) uses 3 seeds per data fraction.

Spherical MNIST. All results in Table 8 report mean $\pm$ std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is measured by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and computing the standard deviation of accuracy, yielding 30 measurements per model (10 rotations $\times$ 3 seeds).

F.5 Dataset licenses

- **PCam**: CC0 1.0 Universal (public domain).
- **OrganMNIST3D**: CC BY 4.0.
- **Spherical MNIST**: derived from MNIST, which is publicly available for research use.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction list four contributions (selective bispectrum theory, completeness proof, `nn.Module` library, and experiments on three benchmarks) that are substantiated in Sections 3–4 and Appendices A–B.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 6 contains a dedicated Limitations paragraph discussing the $\ell = 2$ phase-recovery open question and the octahedral inversion gap. Appendix A includes a "Limitations and open questions" subsection.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Theorem 1 states all assumptions (genericity, gauge-fixing) and a proof sketch appears in Section 3.4. The complete proof with 6 lemmas and 4 propositions is in Appendix A. The octahedral inversion gap analysis is in Appendix B.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: All hyperparameters are in Appendix F (Table 13), architectures in Appendices D–E, optimizer and scheduler details in Table 13, and random seeds are specified. The code is provided as an anonymous repository.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: An anonymous code repository is provided with the submission. All three datasets (PCam, OrganMNIST3D, Spherical MNIST) are publicly available. Sweep scripts and training code are included.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Sections 4.1–4.3 describe experimental setups. Full hyperparameters are in Table 13 (Appendix F), architecture details in Appendices D–E, data preprocessing and splits in Appendix F.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Tables 5–6 (OrganMNIST3D, Spherical MNIST) report mean $\pm$ standard deviation over 3 seeds. Table 4 (PCam Pareto) uses a single seed, which is noted in Appendix F. The aggregation procedure is described in Appendix F.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix F states that all experiments were run on a single NVIDIA H100 GPU (80 GB). Wall-clock times are available in the code repository.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The work develops mathematical invariants and a software library evaluated on publicly available datasets. No human subjects, private data, or dual-use concerns are involved.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: This is foundational mathematical and software research on group-theoretic invariants. We do not foresee direct negative societal impacts.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The released asset is a deterministic mathematical library for computing group-theoretic invariants. It poses no risk for misuse.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All datasets are cited (PCam: CC0 1.0, OrganMNIST3D: CC BY 4.0, MNIST: public domain for research). Dataset licenses are listed in Appendix F.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The library is released under the MIT license with documentation, API reference, and usage examples. Section 3 describes the design and interface.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: This work does not involve crowdsourcing or human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: This work does not involve human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs are not part of the core methodology of this work.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.